

Noetic Learning Math Contest 2026 Spring - Grade 7

Name: Amishi Rajale

1) What is the remainder when 2026 is divided by 5?

Correct! Answer:

2) In the rectangle below, the length is three times the width. The perimeter of the rectangle is 128 inches. What is the length of the rectangle?



Correct! Answer: inches

3) Camila attended a 3-day math camp. Each day after the first, she solved 3 more problems than the previous day. She solved a total of 51 problems during the camp. How many problems did she solve on the first day?

Correct! Answer: problems

4) What is the greatest whole number that is less than 400 and is divisible by 3, 5, and 7?

X Answer:

Correct Answer:

5) Mrs. Garcia conducted a pet survey in her class of 28 students. She found that 11 students have only dogs, 9 students have only cats, and 3 students have neither dogs nor cats. How many students in her class have both dogs and cats?

Pet Survey	
	Dog ☐
	Cat ☐

Correct! Answer: students

6) In the inequality below, the variable w represents a positive integer. What value does w represent?

$$\frac{1}{4} < \frac{2}{w} < \frac{1}{3}$$

Correct! Answer:

7) There are six points in the picture below. How many different triangles can be drawn using point P and two other points as vertices?



X Answer: different triangles

Correct Answer:

8) Lucia and Maria live 1 mile from the gym. Lucia walks to the gym at a constant speed of 3 miles per hour, and Maria jogs to the gym at a constant speed of 5 miles per hour. They leave home at the same time. How many minutes sooner will Maria arrive at the gym than Lucia?

Correct! Answer: minutes sooner

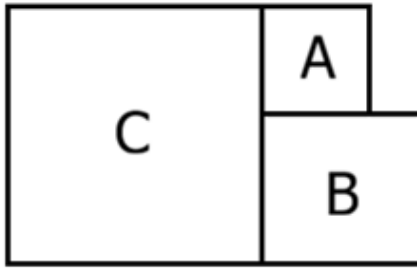
9) Nina's locker number is the greatest 4-digit number whose digits have a product of 90. What is her locker number?

Correct! Answer:

10) There are 60 boys and girls at a party. During a dance, $\frac{1}{3}$ of the boys are paired with $\frac{1}{2}$ of the girls. How many boys are at the party?

Correct! Answer: boys

11) Garvin has three square tiles: A, B, and C. He glued them together to form a new shape, as shown below. The perimeter of tile A is 16 inches. The perimeter of tile B is 24 inches. What is the perimeter of the new shape?



✗ Answer: inches

Correct Answer:

12) In the addition problem below, the same letters represent the same digit, and different letters represent different digits. What digit does the letter A represent?

$$\begin{array}{r} \text{A A} \\ \text{A B} \\ + \text{A B} \\ \hline \text{C C A} \end{array}$$

✗ Answer:

Correct Answer:

13) Penny has bowls, cups, and mugs. All bowls weigh the same, all cups weigh the same, and all mugs weigh the same. She compares their weights using a balance scale and she finds that:

- 2 bowls and 1 mug weigh more than 2 cups and 1 mug.
- 2 mugs and 1 cup weigh more than 2 bowls and 1 mug.

Which item weighs the most?

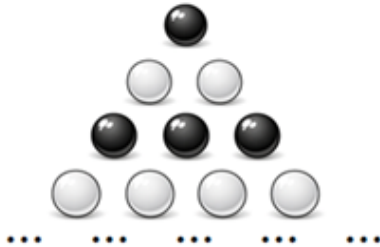
Correct! Answer: ☐ bowl ☐ cup ☒ mug ☐ cannot be determined

14) Caleb, Hussain, and Shyam each have a collection of baseball cards. Hussain has 20% more baseball cards than Caleb. Shyam has 25% more baseball cards than Hussain. Shyam has 180 baseball cards. How many baseball cards does Caleb have?

✗ Answer: baseball cards

Correct Answer:

15) Rohit arranges marbles in rows to form a triangle. The top row contains 1 black marble. Each row below the top row has 1 more marble than the row above it. The rows alternate in color between black and white. When Rohit finishes making his triangle, there are 10 more white marbles than black marbles. How many rows are in his triangle? Note: The picture below shows the top four rows of the triangle.



Correct! Answer: rows

16) It takes Ryan 6 hours to mow a soccer field by himself. It takes Seth 12 hours to mow the same soccer field by himself. Ryan starts mowing the field alone. Then, 3 hours later, Seth joins him. How many more hours will it take them to finish mowing the field together? Note: Ryan and Seth each mow at a constant rate.

Correct! Answer: more hour(s)

17) When the digits of a two-digit number are reversed, the new number is 36 less than the original number. In the original number, how much greater is the tens digit than the ones digit?

Correct! Answer:

18) At Riverview Middle School, two-thirds of the students are in choir and one-half of the students are in band. Every student is in choir, band, or both. If 27 students are in both, what is the total number of students at Riverview Middle School?

Correct! Answer: students

19) Today, $\frac{3}{4}$ of the students at Washington Elementary checked out exactly 1 book each from the school library. The remaining 105 students did not check out any books. The books that were checked out today represent exactly 5% of the total number of books in the library's collection. What is the total number of books in the library's collection?

X Answer: books

Correct Answer:

20) How many cups of an 18% alcohol solution should be added to 1 cup of a 30% alcohol solution to make a solution that is 20% alcohol?

✗ Answer: cups

Correct Answer:

5

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